

UNIVERSITY OF CHINESE ACADEMY OF SCIENCES

DOCTORAL THESIS

The study of Higgs properties in the four-lepton final state at CMS

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Chapter 1

Summary and Outlook

1.1 Summary of the dissertation

After many years of search, ATLAS and CMS collaborations of CERN announced the discovery much awaited Higgs boson in early July 2012 verifying the underlying concept of electroweak symmetry breaking (EWSB) in the SM of particle physics. This triggered essential precise studies of this particle to confirm its nature and to discover some new physics in case of some deviation. With available LHC data, measured properties of Higgs are consistent with SM. Inclusive and differential cross section measurements of Higgs are its very important properties which help to probe its structure and its nature of interaction with decaying particles. These measurements provide a direct probe into SM Higgs sector perturbative QCD calculations at the TeV energy scale.

Measurement of inclusive and differential cross section of the Higgs boson production in proton-proton collisions at $\sqrt{s} = 13$ TeV corresponding to an integrated luminosity of 137.1 fb^{-1} is presented in this dissertation. Measurements are performed by exploiting $H \rightarrow 4\ell$ decay channel ($\ell = e, \mu$), which is known to be a golden channel for such measurements. To minimize the dependence of the measurements on the underlying model of the Higgs boson production mechanism and its properties, the cross sections are extracted in a fiducial volume that closely matches with experimental acceptance. Remaining model dependence is extracted by repeating the measurement for SM and other exotic model and reported as a distinct systematic uncertainty. All measurements include the correction for the finite detection efficiency and resolution effects to allow for direct comparison with the SM predictions. As $\sigma \times BR$ strongly depend upon Higgs mass, the measurements have been extracted assuming $m_H = 125.09 \text{ GeV}$, as reported by CMS experiment [1] from combined measurement in the $H \rightarrow 2\gamma$ and $H \rightarrow 4\ell$ decay channels.

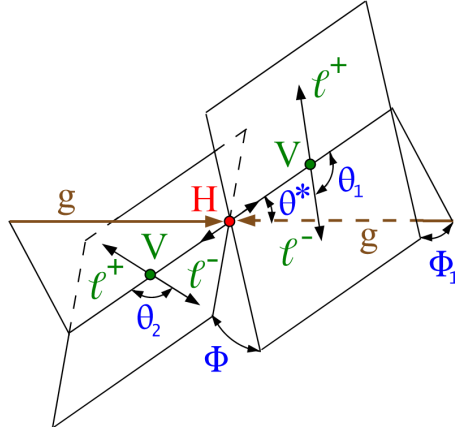


FIGURE 1.1: Higgs event signature

The differential fiducial cross sections are presented as a function of kinematic observables which are sensitive to Higgs production mechanism: rapidity and transverse momentum of Higgs, associated jet multiplicity and transverse momentum of the leading jet. Full Run 2 data (137.1 fb^{-1}) enabled us to explore the differential observables in more fine bins than presented in previous measurements on this topic.

The integrated fiducial cross section is measured to be $2.73^{+0.30}_{-0.29} = 2.73^{+0.23}_{-0.22}(\text{stat.})^{+0.24}_{-0.19}(\text{syst.}) \text{ fb}$ at 13 TeV and SM predicts this to be $2.76^{+0.14}_{-0.14}$. The agreement between SM predictions and measured $H \rightarrow 4\ell$ cross section measurement at $\sqrt{s} = 13 \text{ TeV}$ is found to be good. These model independent cross section measurements can be compared with any theoretical predictions in future with better accuracy.

1.2 Outlook

The presented measurements of Higgs differential cross sections are related to Higgs production. There are certain things that can be done which are beyond the scope of this dissertation:

- With more data, get more precise results
- Study more differential observables
- Include additional theoretical prediction(s) for comparison with the observed cross section in differential bins

- Include potential theory interpretations for differential observables

The current measurements are emerged with significant statistical uncertainties. The studies based on simulations show that the uncertainties on Higgs cross section measurements with 300 fb^{-1} pp collisions data (as expected from LHC Run 3) would be significantly less than the current theoretical uncertainties. More data will help to affirm the SM Higgs boson if measurements be in agreement with the SM expectations, otherwise any deviation from SM predictions will open the door to some new physics.

Studying more differential observables are also important from different aspects. There are more *production* and as well as *decay* related observables ($\theta^*, \theta_1, \theta_2, \Phi, \Phi_1$ as shown in Figure. 1.1) which can be studied with currently available data from CMS [event_signature]. With respect to earlier iterations, current data would allow to exploit more differential bins for a particular observable. Moreover, with more statistics there is opportunity to study the double differential cross section (as function of two observables). In $H \rightarrow 4\ell$, there are ongoing efforts for additional differential measurements alongwith potential theory interpretations; however, it is yet to accomplish to be made public. Preliminary version of some of the new differential measurements have been shown in Appendix. In the reported measurements, the observed cross sections are compared with respective theoretical predictions based on POWHEG and NNLOPS; however, including additional theoretical prediction(s) e.g. MADGRAPH5 are also underway to be part of next manuscript.

91 Bibliography

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93 and tests of compatibility of its couplings with the standard model predictions
94 using proton collisions at 7 and 8 TeV”. In: *Eur. Phys. J. C* 75 (2015), p. 212.